LESSON 01 / 05

## Differentiate a product n times without expanding every case

Leibniz's rule generalizes the product rule. For sufficiently differentiable $u, v$,

$$(uv)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} u^{(k)} v^{(n-k)}.$$

Here $u^{(0)} = u$. The formula is especially short when one factor is a polynomial, because sufficiently high derivatives of that factor vanish.

**WORKED EXAMPLE**

**Only three Leibniz terms survive for x^2e^x**

Take $u = x^2$ and $v = e^x$. Since $u''' = 0$ and every derivative of $v$ is $e^x$, for $n \geq 2$,

$$(x^2 e^x)^{(n)} = e^x[x^2 + 2nx + n(n-1)].$$

In particular,

$$(x^2 e^x)'' = e^x(x^2 + 4x + 2), \qquad (x^2 e^x)''' = e^x(x^2 + 6x + 6).$$

For $n = 0, 1$, direct substitution gives $x^2 e^x$ and $e^x(x^2 + 2x)$, also consistent with the displayed polynomial formula.

**WORKED EXAMPLE**

**The same three-term pattern differentiates x^2 cos x**

For $n \geq 2$, Leibniz's rule gives

$$D^n(x^2 \cos x) = x^2 \cos\left(x + \tfrac{n\pi}{2}\right) + 2nx \cos\left(x + \tfrac{(n-1)\pi}{2}\right) + n(n-1)\cos\left(x + \tfrac{(n-2)\pi}{2}\right).$$

Only derivatives of $x^2$ through order $2$ survive. The cases $n = 0, 1$ are checked directly; the formula also reduces correctly for $n = 1$.

## Recognize when a polynomial terminates the sum

If $p$ has degree $m$, then $p^{(k)} = 0$ for $k > m$. Hence $(pv)^{(n)}$ needs only $0 \leq k \leq \min(n, m)$. This is a structural simplification, not an assumption that $v$ also has terminating derivatives.

## A reliable higher-derivative workflow




READ THE FIGURE

*Most errors come from omitted binomial coefficients or from assigning the same derivative order to both factors.*

COMMON TRAP

**Leibniz's rule does not say (uv)^(n)=u^(n)v+uv^(n)**

*Those are only the two endpoint terms. Unless all intermediate terms genuinely vanish, the full binomial sum is required.*

LESSON 02 / 05

## Powers and reciprocals have compact nth derivatives

For a real exponent $r$ and a domain on which $ax+b>0$,

$$D^n(ax+b)^r = a^n r(r-1)\cdots(r-n+1)(ax+b)^{r-n}.$$

If $m$ is a nonnegative integer and $0 \le n \le m$, this becomes

$$D^n(ax+b)^m = \frac{m!}{(m-n)!}a^n(ax+b)^{m-n},$$

and derivatives of order greater than $m$ vanish. For $a \neq 0$ and $ax+b \neq 0$,

$$D^n \frac{1}{ax+b} = \frac{(-1)^n n!\, a^n}{(ax+b)^{n+1}}.$$

Thus the fourth derivative of $(2x+1)^3$ is $0$, while the second derivative of $1/x$ is $2/x^3$ on either side of its domain. In particular,

$$D^n \frac{1}{1-x} = \frac{n!}{(1-x)^{n+1}} \qquad (x \neq 1).$$

## Split a logarithm's derivative without inventing real logarithms

For $y = \ln(x^2-1)$, the real domain is $|x|>1$. On either component,

$$y' = \frac{2x}{x^2-1} = \frac{1}{x-1} + \frac{1}{x+1}.$$

Therefore, for $n \ge 1$,

$$y^{(n)} = (-1)^{n-1}(n-1)!\,[(x-1)^{-n} + (x+1)^{-n}].$$

The decomposition is applied to the **derivative** as a rational identity. On $x<-1$, the original logarithm is real, but $\ln(x-1)$ and $\ln(x+1)$ are not separately real.

## Exponential-trigonometric derivatives rotate by a fixed phase

For $(a,b) \neq (0,0)$, let $r = \sqrt{a^2+b^2} > 0$ and choose $\phi$ with $\cos\phi = a/r$, $\sin\phi = b/r$. Then

$$D(e^{ax}\cos bx) = re^{ax}\cos(bx+\phi).$$

Iterating gives

$$D^n(e^{ax}\cos bx) = r^n e^{ax}\cos(bx+n\phi).$$

For $a=b=1$, this is

$$D^n(e^x \cos x) = 2^{n/2} e^x \cos\left(x + \frac{n\pi}{4}\right).$$

WORKED EXAMPLE

**A differential identity can replace repeated messy expansion**

For $y=\sqrt{1-x^2}$ on $|x|<1$,

$$y'=-\frac{x}{y}, \qquad y''=-\frac{1}{y^3}.$$

Consequently

$$(1-x^2)y''-xy'=-\frac{1-x^2}{y^3}+\frac{x^2}{y}=-\frac{1}{y}+\frac{x^2}{y}=\frac{x^2-1}{y}=-y.$$

This exact identity is a useful independent check of both derivatives.

## Choose a closed form from the expression's structure

| Structure | Closed-form cue | Caution |
|---|---|---|
| Polynomial power | Falling factorial | degree and order |
| Real power | Falling factorial | positive base |
| Reciprocal | Sign and n! | exclude pole |
| Log of quadratic | Split its derivative | \|x\| above 1 |
| Exp. times sine | Amplitude; phase | choose phi |
| Implicit radical | Use exact relation | check domain |

READ THE FIGURE

Closed forms shorten iteration, but a low-order direct derivative remains the fastest way to detect a missing factor.

LESSON 03 / 05

## First exclude degenerate solutions, then differentiate

Suppose a differentiable $f:\mathbb{R}\to\mathbb{R}$ satisfies

$$f(x+y)=f(x)f(y)$$

and $f$ is not identically zero. Since $f(0)=f(0)^2$, $f(0)$ is $0$ or $1$. If $f(0)=0$, then $f(x)=f(x)f(0)=0$ for every $x$, which is excluded. Hence $f(0)=1$.

Also $f(x)=f(x/2)^2\geq 0$. If $f(a)=0$ for some $a$, then $f(x)=f(a)f(x-a)=0$ for every $x$, again impossible. Thus $f>0$ everywhere.

Differentiate the functional equation with respect to $y$ and set $y=0$:

$$f'(x+y)=f(x)f'(y)\quad\Longrightarrow\quad f'(x)=cf(x),\qquad c=f'(0).$$

Now $g(x)=e^{-cx}f(x)$ has derivative $e^{-cx}(f'-cf)=0$. The zero-derivative theorem, justified in Extension V, gives $g(x)=g(0)=1$. Therefore

$$f(x)=e^{cx},$$

and direct substitution verifies every such function.

## A product argument produces logarithms

Suppose instead that $f:(0,\infty)\to\mathbb{R}$ is differentiable and

$$f(xy)=f(x)+f(y).$$

Putting $x=y=1$ gives $f(1)=0$. Differentiate with respect to $y$:

$$xf'(xy)=f'(y).$$

Set $y=1$ and $a=f'(1)$ to obtain $xf'(x)=a$, hence $f'(x)=a/x$. Since $f(x)-a\ln x$ has zero derivative and vanishes at $1$,

$$f(x)=a\ln x\qquad(x>0).$$

Again, substitution verifies the whole family.

## An additive equation has constant derivative

If a differentiable $f:\mathbb{R}\to\mathbb{R}$ satisfies

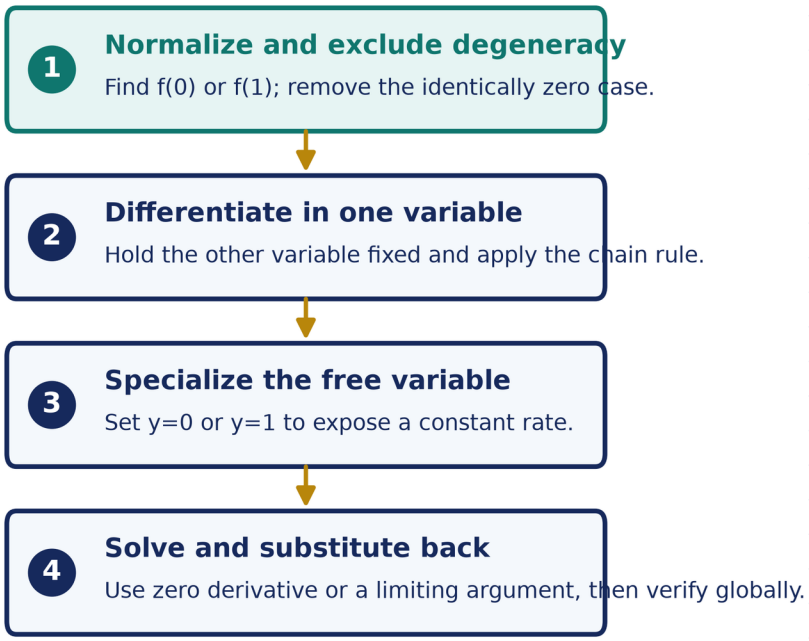
$$f(x+y)=f(x)+f(y),$$

then $f(0)=2f(0)$, so $f(0)=0$. Differentiate with respect to $y$ and set $y=0$:

$$f'(x+y)=f'(y)\quad\Longrightarrow\quad f'(x)=f'(0)=a.$$

Thus $f(x) - ax$ has zero derivative and vanishes at $0$, giving $f(x) = ax$. For the source value $a = f'(0) = 3$, the answer is $f(x) = 3x$. Differentiability is what rules out nonregular additive solutions.

### Differentiation converts functional equations into rate equations




**READ THE FIGURE**
*The exponential case excludes zero before solving; the product-argument case is defined only on positive reals. Additive equations need no logarithm.*

**COMMON TRAP**
***Differentiability is an essential hypothesis***
Without regularity, additive and related functional equations can admit pathological solutions. Differentiation cannot be applied merely because an algebraic relation looks familiar.

LESSON 04 / 05

## A derivative can exist everywhere yet fail to be continuous

Define $f(0) = 0$ and, for $x \neq 0$,

$$f(x) = x^2 \sin(1/x).$$

At the origin,

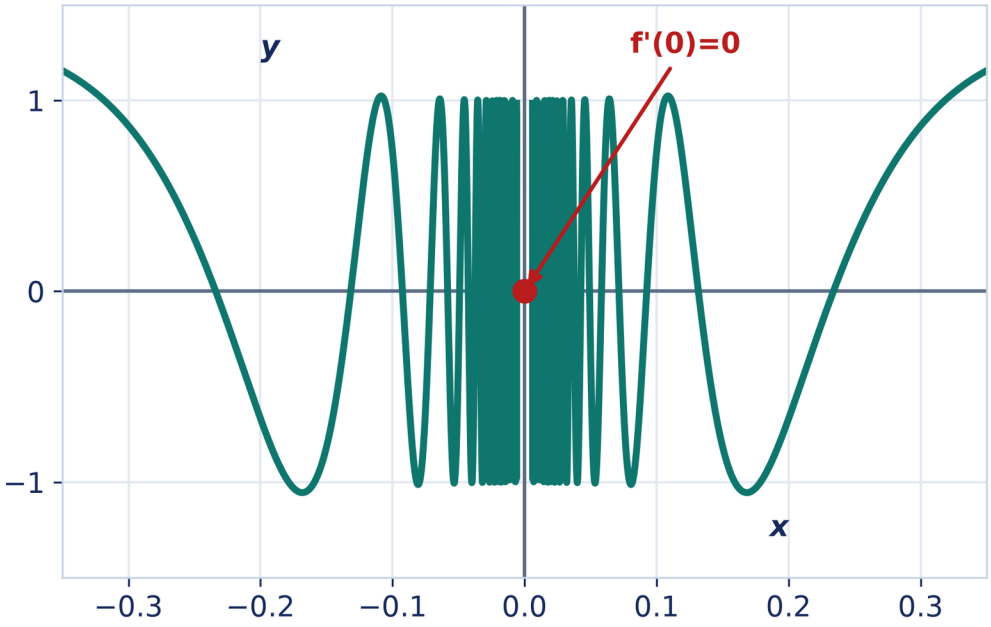
$$\left|\frac{f(x) - f(0)}{x}\right| = |x\sin(1/x)| \leq |x| \to 0,$$

so $f'(0) = 0$. Away from the origin,

$$f'(x) = 2x\sin(1/x) - \cos(1/x).$$

Along $x_n = 1/(2n\pi)$, this equals $-1$; along $y_n = 1/((2n+1)\pi)$, it equals $1$. Both input sequences tend to $0$, so $f'(x)$ has no limit at $0$. Thus $f'$ exists everywhere but is discontinuous there.

**This everywhere-existing derivative oscillates near zero**




READ THE FIGURE
The plotted formula applies away from 0; the marked value at 0 comes from the difference quotient. No single limiting value matches the oscillating branches.

## Darboux's theorem still forces intermediate values

If $f$ is differentiable on an interval, then $f'$ has the **intermediate value property**: between any two attained derivative values, every intermediate value is also attained. Continuity of $f'$ is not required.

Consequently, a genuine derivative cannot have a simple jump discontinuity: values immediately on either side could not skip an entire interval of intermediate slopes. The example above instead has an oscillatory discontinuity, which is compatible with Darboux's theorem.

KEY RESULT

**Separate three statements**

Differentiability of $f$ does not imply continuity of $f'$. Continuity of $f'$ is stronger. Darboux's intermediate value property follows from differentiability alone and rules out jump discontinuities, but not oscillation.

LESSON 05 / 05

## Rolle's theorem turns equal endpoints into a stationary point

If $f$ is continuous on $[a,b]$, differentiable on $(a,b)$ and $f(a)=f(b)$, then there is $c\in(a,b)$ with $f'(c)=0$. Continuity on the closed interval gives an extreme value. If every extreme occurs only at endpoints, equal endpoint values force $f$ to be constant; otherwise an interior extreme has zero derivative.

**WORKED EXAMPLE**

**Two Rolle points for x^3-x on [-1,1]**

Here $f(-1)=f(1)=0$ and

$$f'(x)=3x^2-1.$$

Thus $f'(c)=0$ exactly at $c=\pm 1/\sqrt{3}$, both inside $(-1,1)$.

## The mean value theorem tilts the same picture

Under the same continuity and differentiability hypotheses, but without equal endpoints,

$$\frac{f(b)-f(a)}{b-a}=f'(c)$$

for some $c\in(a,b)$. To prove this, apply Rolle's theorem to

$$g(x)=f(x)-f(a)-\frac{f(b)-f(a)}{b-a}(x-a),$$

which has $g(a)=g(b)=0$. For $f(x)=x^2$ on $[1,3]$, the secant slope is $4$, so $2c=4$ and $c=2$.

## MVT can prove global inequalities without optimization

For $a\neq b$, MVT gives $c$ between $a$ and $b$ such that

$$\sin a-\sin b=(a-b)\cos c.$$

Hence $|\sin a-\sin b|\le|a-b|$. Taking $a=x\ge 0$ and $b=0$ also gives $\sin x\le|\sin x|\le x$.

Similarly, for $x\neq 0$, MVT gives $c$ between $0$ and $x$ with $e^x-1=xe^c$. If $x>0$, then $e^c>1$; if $x<0$, then $e^c<1$ and multiplication by the negative $x$ reverses the inequality. In both cases $e^x-1>x$, while equality occurs at $x=0$. Thus $e^x\ge 1+x$ for every real $x$, with equality only at $0$.

For $x>1$, applying MVT to $\ln x$ on $[1,x]$ gives $c\in(1,x)$ with

$$\ln x=\frac{x-1}{c}<x-1.$$

For $0 < x < 1$, there is $c \in (x, 1)$ such that $\ln x = (x-1)/c < x - 1$; the negative numerator and $0 < c < 1$ preserve the stated strict inequality. At $x = 1$, equality holds. Hence

$$\ln x \le x - 1 \qquad (x > 0).$$

**The logarithm lies below its tangent upper bound x-1**




READ THE FIGURE

The two graphs meet at (1,0) and agree in slope there. MVT proves the full inequality; the figure is an independent geometric check, not the proof.

COMMON TRAP

**This extension is a bridge, not Chapter 19**

Rolle and MVT are included because the source introduces them beside derivatives. Systematic monotonicity, extrema, optimization and advanced tangent-normal applications belong to Chapter 19 — Applications of Derivatives.